

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Computer Science)	Semester:	Spring 2026
	Duration:	15 minutes	Total Marks:	15
	Paper Date:		Section	6B1
	Exam Type:	Quiz 1 - Chapter 1	Page(s):	2

Student Name

Roll No.

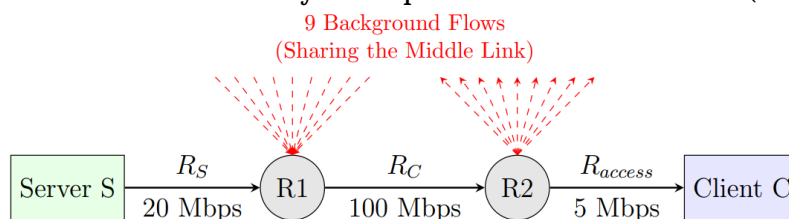
Section:

Q1. Carefully read the Question before attempting:

[5 marks]

Consider the file transfer scenario shown below. Server S sends a file to Client C through two intermediate routers R1 and R2.

- Link S → R1: Rate $R_S = 20$ Mbps.
- Link R1 → R2: Shared link, Rate $R_C = 100$ Mbps. This link is shared by 9 other background traffic flows (assume all are working at full capacity).
- Link R2 → C: Rate $R_{access} = 5$ Mbps.
- The 9 background flows consume exactly 10 Mbps each of the middle link (R1 → R2).



Calculate the throughput for the file transfer from S to C. Show the available bandwidth for each link and identify the bottleneck link.

Solution:

- Background consumption on middle link = $9 \times 10 \text{ Mbps} = 90 \text{ Mbps}$
- Available bandwidth on middle link = $100 \text{ Mbps} - 90 \text{ Mbps} = 10 \text{ Mbps}$

Available bandwidth on each link

- S → R1: 20 Mbps
- R1 → R2 (available): $100 - 90 = 10 \text{ Mbps}$
- R2 → C: 5 Mbps

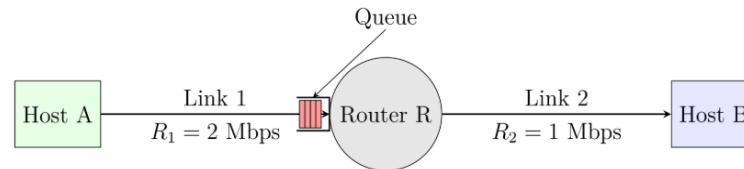
Throughput = minimum available along path = $\min(20, 10, 5) = 5 \text{ Mbps}$.

Bottleneck link: R2 → C (access link) at **5 Mbps**.

Q2. Carefully read the Question before attempting:**[10 marks]**

Consider a packet-switched network where Host A sends a packet to Host B via a single Router R.

- Link 1 (A → R): Transmission Rate $R_1 = 2$ Mbps. Propagation Delay $d_{\text{prop1}} = 10$ ms.
- Link 2 (R → B): Transmission Rate $R_2 = 1$ Mbps. Propagation Delay $d_{\text{prop2}} = 10$ ms.
- Packet Size: All packets have a length $L = 2000$ bits.
- Processing Delay: Negligible at Router R ($d_{\text{proc}} \approx 0$).
- Queuing: Router R has a buffer. Since $R_1 > R_2$, packets arriving at R faster than they can be transmitted to B, may cause a queue to build up.



1. Calculate the transmission delays at each Link?
2. What is the total end to end delay considering negligible queuing and processing delays (router must receive the whole packet before starting transmission)?

Solution:

Link 1 (A → R):

$dt1 = 2000 \text{ bits} / 2,000,000 \text{ bits/s}$
Divide: $2000 \div 2,000,000 = 0.001 \text{ s}$
Convert to ms: $0.001 \text{ s} \times 1000 = 1 \text{ ms}$
So $dt1 = 1 \text{ ms}$

Link 2 (R → B):

$dt2 = 2000 \text{ bits} / 1,000,000 \text{ bits/s}$
Divide: $2000 \div 1,000,000 = 0.002 \text{ s}$
Convert: $0.002 \text{ s} \times 1000 = 2 \text{ ms}$
So $dt2 = 2 \text{ ms}$

Total End-to-End Delay (store-and-forward, negligible queuing & processing):

End-to-End Delay = $dt1 + d_{\text{prop1}} + dt2 + d_{\text{prop2}}$
 $= 1 \text{ ms} + 10 \text{ ms} + 2 \text{ ms} + 10 \text{ ms}$
 $= 23 \text{ ms}$